Geometric Observables for Financial Regime Detection

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Abstract

We extract four geometric observables—Berry Phase Rate, Spectral Entropy, Reduced State Purity, and Hamiltonian Sensitivity—from a learned spectral embedding of equity-index returns and evaluate them as regime-shift detectors against 46 classical and machine-learning baselines on 17 historical crises spanning 2000–2024. Under walk-forward nested hyperparameter selection on nine labelled windows, the Berry Phase Rate achieves an unbiased out-of-sample median Cohen's $d = 0.72$ (95% percentile-bootstrap CI $[0.34, 1.18]$, 10,000 resamples) and produces approximately 67% fewer false alarms per year than a label-supervised Random Forest (1.2 vs. 3.6 per year). Reduced State Purity attains the highest in-sample separability of any method ($d = 0.83$), tied closely by the Absorption Ratio [1] ($d = 0.80$); geometric and classical channels are largely uncorrelated ($|\rho| \approx 0.13$), suggesting they capture distinct risk signals. Score construction is unsupervised; hyperparameter selection is the only supervised step.

Keywords: regime detection, financial crises, walk-forward validation, false-alarm rate, Berry curvature, Fubini–Study metric, quantum Fisher information, spectral metric learning, geometric observables, QCML

1. Introduction

Financial markets switch between calm trends and violent drawdowns, sometimes within a single trading day. Detecting these regime transitions is a core problem in quantitative finance. Hidden Markov Models [2, 3], Bayesian Online Changepoint Detection [4], and supervised classifiers all work in specific settings, but they are defined on flat Euclidean feature spaces. An alternative is to measure the geometry of the data itself.

Quantum Cognitive/Cognition Machine Learning (QCML; both appear in the literature) [5, 6] embeds data as unit vectors in a finite-dimensional complex space and represents features as Hermitian operators. The construction equips the data manifold with a learned metric, a curvature, and a spectral gap—geometric quantities that respond to deformations in the data-generating process (Section 2.1). Everything runs on classical hardware; "quantum" refers to the Hilbert-space formalism, not quantum computing.

The QCML embedding yields four geometric observables, Berry Phase Rate, Spectral Entropy, Reduced State Purity, and Hamiltonian Sensitivity, which we compare against classical baselines including Absorption Ratio, Hamilton Markov-Switching, and GARCH (Table 2).

The primary evaluation is *crisis-window separability*— does the score distribution during a known crisis differ significantly from a non-crisis period? This is an offline event study, measuring sensitivity rather than real-time prediction. Walk-forward evaluation (Section 5.2) tests causal detection with strictly past-fitted preprocessing.

Supervision level. Score construction is unsupervised (Algorithm 1); the pipeline is *semi-supervised* only because hyperparameters are selected against crisis labels (Section 5.3).

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Contributions.

1. **Walk-forward detection.** Nested HPO yields a Berry Phase Rate OOS median $d = 0.72$ (CI $[0.34, 1.18]$), the highest per-crisis Cohen's d among three walk-forward detectors on five of nine crises, with $\sim$67% fewer false alarms than Random Forest (1.2 vs. 3.6 per year) and no crisis labels.
2. **QCML geometric framework.** Four geometric observables from a single embedding, each probing a different property of the data manifold (curvature, entropy, sensitivity, entanglement).
3. **Offline separability.** On 17 crises (2000–2024), Reduced Purity ($d = 0.83$) ranks first of 46 methods, with Absorption Ratio ($d = 0.80$) as the classical benchmark; the remaining geometric channels ($d =$ 0.53–0.61) contribute orthogonal crisis coverage.
4. **Risk-overlay viability.** Out-of-sample false-alarm rates of $\sim$1.2/yr for the Berry channel are low enough to support a practitioner risk-management overlay; a SPY de-risking experiment (Section 6.3) cuts maximum drawdown by $\sim$51% with no degradation in Sharpe ratio.

Scope and limitations. Offline rankings measure crisis-window sensitivity, not real-time detection. Reduced Purity ($d = 0.83$ offline) is sensitive to bipartition choice and drops to $d \approx 0.26$ on frozen holdout, so high offline separability does not guarantee out-of-sample stability; Berry Phase Rate, which ranks lower offline ($d = 0.61$), gives the strongest walk-forward evidence and is the load-bearing OOS claim of this paper. The geometric channels complement, rather than replace, existing methods [7, 8].

1.1. Related Work

Several non-Euclidean frameworks have been applied to financial data (Table 1).

Information geometry and optimal transport. The Fisher–Rao metric [9] has been applied to interest rate manifolds [10] and return distributions [11]; Horvath et al. [12] cluster regimes via Wasserstein distance. These provide a metric on the data manifold, nothing more. There is no curvature, no holonomy, and no spectral gap to work with.

TDA and network curvature. The closest curvature-based comparator is Sandhu et al. [13], who put Ollivier–Ricci curvature on correlation networks. The Berry curvature used here is different in kind: it lives on the data manifold, not an inter-asset graph, and it produces integer Chern invariants via Chern–Weil theory. Two topology-based approaches are closer in spirit: Gidea and Katz [14] track persistence landscapes as crash early-warnings, and Guritanu et al. [15] use persistent homology for causal crisis detection. Persistent-homology features are complementary to differential-geometric observables like Berry curvature and the Fubini–Study metric, not substitutes.

Random matrix theory and signatures. Laloux et al. [16] separate signal from noise via Marchenko–Pastur. That analysis cares about a noise edge; our spectral gap is an order parameter for a phase transition— a stronger claim about what the eigenvalue structure encodes. On the signature side, Issa and Horvath [17] run MMD tests on rough-path signatures, and Bronstein et al. [18] impose geometric priors instead of reading the geometry off the data.

Classical regime detection. Hamilton's [2] Markov-switching, Ang and Bekaert's [3] financial extensions, and Bollerslev's [19] GARCH are all included as baselines.

QCML literature. The framework itself is from Candelori et al. [5] and Abanov et al. [6]. Financial applications have been sketched in industry papers [20, 21]. We are not proposing a new framework; we are taking theirs, pulling four geometric observables off a single embedding, and putting them through a statistically serious comparison against established baselines with strict walk-forward validation.

Table 1: Non-Euclidean approaches to financial regime detection. Columns indicate which geometric tools each framework provides.

Framework	Metric	Curvature	Gauge	Topol. Invt.	Spect. Gap
Info. Geometry	Fisher–Rao	Sectional	α-conn.	—	—
Optimal Transport	Wasserstein	—	—	—	—
TDA	—	—	—	Betti	—
Network Ricci	—	Ollivier	—	—	—
Random Matrix Thy.	—	—	—	—	MP edge
Rough Path Sig.	Sig. kernel	—	—	—	—
QCML (ours)	**Fubini–Study**	**Berry**	**U(1)**	**Chern**	$\boldsymbol{\Delta(x)}$

2. QCML Framework

We work in a finite-dimensional Hilbert space $\mathcal{H} \cong \mathbb{C}^n$. A *state* is a unit vector $|\psi\rangle \in \mathcal{H}$, and an *observable* is a Hermitian operator— $A = A^\dagger$—that is, an operator that is equal to its adjoint [5, 6].

Given $x = (x_1, \ldots, x_p) \in \mathbb{R}^p$ and Hermitian operators $\{A_k\}_{k=1}^p$, the QCML *error Hamiltonian* [5, 20] is:

$$H(x) = \frac{1}{2} \sum_{k=1}^{p} (A_k - x_k\, I)^2\,, \tag{1}$$

where I is the $n \times n$ identity. The *quasi-coherent state* at x is the ground state:

$$|\psi(x)\rangle = \underset{|\phi\rangle,\, \|\phi\|=1}{\arg\min}\, \langle \phi | H(x) | \phi \rangle\,. \tag{2}$$

The pullback of the Fubini–Study metric defines the *quantum metric tensor* [22]:

$$g_{ab}(x) = \mathrm{Re}\, \langle \partial_a \psi | \partial_b \psi \rangle - \langle \partial_a \psi | \psi \rangle \langle \psi | \partial_b \psi \rangle\,, \tag{3}$$

where indices $a, b \in \{0, \ldots, p-1\}$ run over PCA components, and the *Berry curvature* [23, 6]:

$$F_{ab}(x) = -2\, \mathrm{Im}\, \langle \partial_a \psi | \partial_b \psi \rangle\,. \tag{4}$$

Equation (4) holds in the parallel-transport gauge $\langle \psi | \partial_a \psi \rangle = 0$; our numerical implementation uses the gauge-invariant plaquette formula (Section 3.1).

Theorem 1 (Smoothness [5])**.** *If the spectral gap $\Delta(x) = E_1(x) - E_0(x) > 0$ on an open set $U \subset \mathbb{R}^p$, then $x \mapsto |\psi(x)\rangle$, $g_{ab}(x)$, and $F_{ab}(x)$ are C^∞ on U.*

Proof sketch. By the implicit function theorem applied to the eigenvalue equation $H(x)|\psi\rangle = E_0|\psi\rangle$, $\langle \psi | \psi \rangle = 1$; the non-degenerate spectral gap ensures invertibility of the relevant Jacobian. See Appendix Appendix A for the full argument. □

Operator construction. We use *PCA-inspired* operators: Pauli-basis matrices scaled by PCA eigenvalues, $A_k = \sqrt{\lambda_k}\, \sigma_k$. An alternative is *random* Hermitian operators: $A = (M + M^\dagger)/2$, where entries of M are drawn from a standard complex normal distribution. Random operators avoid the near-degeneracy that PCA-inspired operators can produce for certain Hilbert dimensions (see Section 5.3). All stochastic elements (random operator generation, bootstrap resampling, RF training) use seed 42 for reproducibility. An ablation comparing random, PCA-inspired, and learned-scaling operators appears in Section 5.3. Full QCML gradient-descent training [20, 21] is left for future work.

Why fixed operators suffice for detection. The QCML framework [5] normally learns operators via reconstruction loss to estimate intrinsic dimension and generate predictions. Detection asks less. By Theorem 1, any operator set with a positive spectral gap $\Delta(x) > 0$ induces a C^∞ geometry with a well-defined metric, curvature, and topological invariants; regime shifts perturb $H(x)$ and the induced geometry responds, whether or not the operators were ever optimized. The ablation (Section 5.3) is blunt about this: heuristic random and PCA-inspired operators already achieve $d > 0.5$ with no gradient training, and discriminative operator learning does not systematically improve on them.

Proposition 2 (Fisher information interpretation)**.** *For the pure states $|\psi(x)\rangle$ arising as ground states of $H(x)$, the quantum metric tensor* (3) *satisfies $4\,g_{ab}(x) = [F_Q]_{ab}(x)$, where F_Q is the quantum Fisher information matrix [24, 25, 26]. Consequently:*

(i) The QFI pseudo-determinant $\det^+(g)$ is proportional to the volume element of the Fisher information metric on the statistical manifold: $\det^+(g) = 4^{-r}\det^+(F_Q|_r)$, where $r = \mathrm{rank}(g)$.

(ii) The quantum Cramér–Rao bound [27, 28] $\mathrm{Var}(\hat{x}_a) \geq \frac{1}{4}[g^{-1}]_{aa}$ implies that the quantum metric controls the ultimate precision for estimating data coordinates from state measurements.

(iii) Near regime boundaries where $|\psi(x)\rangle$ changes rapidly, $\det^+(g)$ increases, reflecting increased distinguishability of neighboring data distributions: $D_{\mathrm{KL}}(P_x \| P_{x+dx}) \approx 2\,dx^T g\,dx$.

Parts (i)–(iii) follow from standard results in quantum estimation theory; see Appendix Appendix A.

2.1. Intrinsic Manifold Dimension

Why the Fubini–Study pullback metric $g_{ab}(x)$? Candelori et al. [5] give a concrete answer: the eigenvalue spectrum of g_{ab} has a spectral gap at position d equal to the intrinsic dimension of the data manifold. MLE, DANCo, and TwoNN estimators create "shadow dimensions" from noise artifacts. The pullback metric does not, because it reads dimension off a spectral gap that is stable under noise. Our geometric observables measure deformations of a manifold whose metric $g(x)$ already reflects the underlying data geometry, tying the construction to classical results in random matrix theory [16, 29] and factor models [30].

Empirical demonstration. We compute the eigenvalue spectrum of $g_{ab}(x_t)$ on the SPY/DIA feature space at each time step and identify the spectral gap. In normal periods, the largest eigenvalue ratio $\lambda_2/\lambda_3 \approx 4.5$ indicates a sharp gap at $d \approx$ 2–3, consistent with Laloux et al. [16, 29] ($\sim$6% signal eigenvalues) and Bai and Ng [30] ($\sim$2 latent factors). During crises, this ratio drops to $\sim$2.8 and the metric participation ratio $\mathrm{PR}(g) = (\sum_i \lambda_i)^2/\sum_i \lambda_i^2$ rises from 2.63 ± 0.54 to 2.92 ± 0.57 (Cohen's $d = 0.54$, $p < 10^{-56}$). The manifold *unfolds* as herding breaks the normal low-dimensional structure (Figure 1). The QCML spectral gap dimension detects this shift with $d = 0.69$ ($p < 10^{-104}$), whereas PCA effective rank barely distinguishes regimes ($d = 0.13$), a five-fold sensitivity gap. Spectral Entropy (Section 3.2) tracks the full eigenvalue distribution and achieves mid-range offline separability among geometric channels ($d = 0.53$, rank 12 of 46).

Connection to existing observables. The Absorption Ratio [1] measures the fraction of variance explained by the top eigenvalue of the rolling correlation matrix. That is a *linear* dimension measure: PCA eigenvalue concentration, nothing more. The Fubini–Study pullback generalizes it nonlinearly. The QCML embedding sends data through $x \mapsto |\psi(x)\rangle$ into $\mathbb{CP}^{n-1}$, and the pullback metric picks up curvature and topological structure that flat PCA space simply does not see. Candelori et al. [5] show that gradient-descent-trained operators give the cleanest manifold approximation; our fixed operator basis preserves the spectral-gap structure that actually matters for detection, with the caveat that absolute dimension estimates are only approximate.

3. Geometric Observables

Four geometric observables are extracted from the QCML embedding, each computed at every time step t and converted to z-scores via Algorithm 1 (Table 2). Absorption Ratio [1] is included as a classical baseline to calibrate the value added by geometric structure.

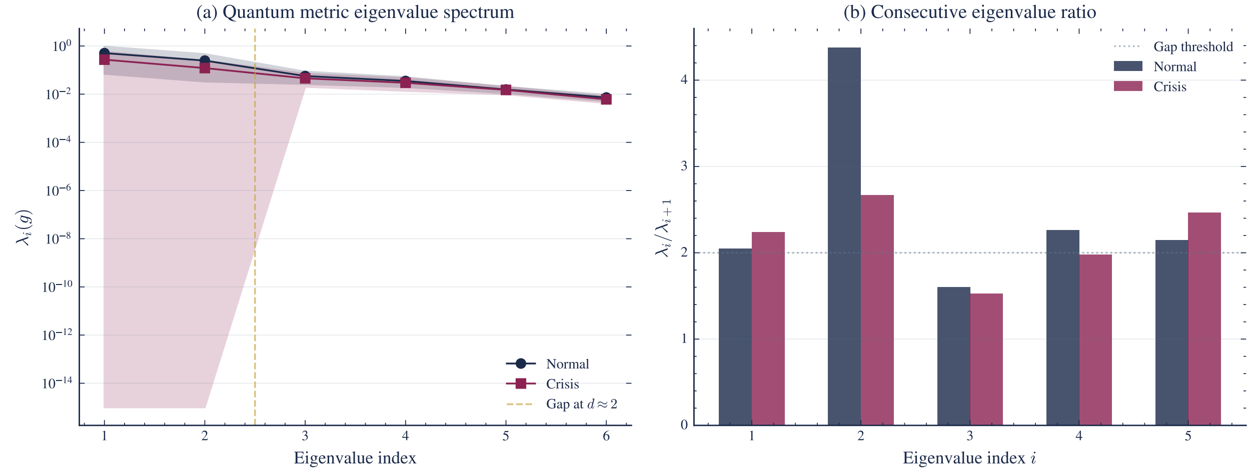

Figure 1: Quantum metric eigenvalue spectra at normal vs. crisis timepoints. (a) Mean spectrum with $\pm 1\sigma$ bands; the dashed line marks the spectral gap at $d \approx 2$. (b) Consecutive eigenvalue ratios λ_i/λ_{i+1}; the sharp gap at $i = 2$ in normal periods (ratio ≈ 4.5) weakens during crises (ratio ≈ 2.8), indicating manifold dimension increase.

Table 2: Geometric observatory channels studied in this paper.

Channel	Family	Geometric property	Equation
Berry Phase Rate	Holonomy	U(1) holonomy change rate	(5)
Spectral Entropy	Spectral	Excitation distribution	(7)
Hamiltonian Sensitivity	State	Perturbation variance	(8)
Reduced Purity	State	Subsystem separability	(9)

3.1. Berry Phase Rate

The Berry Phase Rate is the absolute increment of Berry curvature between consecutive time steps:

$$\dot{\gamma}(t) = |F_{01}(x_t) - F_{01}(x_{t-1})| \,, \tag{5}$$

where F_{01} is the Berry curvature (4) in PCA directions $(0, 1)$, computed via the plaquette (Wilson loop) method.

Hyperparameters. Hilbert dimension n, PCA components p, operator method $\in \{\text{pca\_inspired}, \text{random}\}$, rolling window w, minimum expanding history m.

Plaquette computation. The Berry curvature F_{ab} is computed via the discrete plaquette (Wilson loop) method. For each pair of PCA directions (a, b) at point x:

$$\begin{aligned} F_{ab}(x) = -\frac{1}{\epsilon^2}\, \mathrm{Im}\, \log\big(&\langle\psi(x)|\psi(x{+}\epsilon e_a)\rangle \\ &\times \langle\psi(x{+}\epsilon e_a)|\psi(x{+}\epsilon e_a{+}\epsilon e_b)\rangle\langle\psi(x{+}\epsilon e_a{+}\epsilon e_b)|\psi(x{+}\epsilon e_b)\rangle\langle\psi(x{+}\epsilon e_b)|\psi(x)\rangle\big), \end{aligned} \tag{6}$$

This gauge-invariant formula avoids explicit phase-fixing [23, 31].

3.2. Spectral Entropy

Shannon entropy of excitation energy weights:

$$S(t) = -\sum_{n=1}^{N-1} w_n \log w_n \,, \qquad w_n = \frac{E_n(t) - E_0(t)}{\sum_{m=1}^{N-1}(E_m(t) - E_0(t))} \,. \tag{7}$$

Low entropy indicates energy concentration in few excitations (ordered regime); high entropy signals a disordered/crisis regime. Among the four geometric channels studied here, Spectral Entropy achieves mid-range offline separability ($d = 0.53$, rank 12 of 46).

3.3. State Dynamics

Hamiltonian Sensitivity. Variance of the Hamiltonian perturbation:

$$\sigma_H^2(t) = \langle\psi(x_t)|\Delta H^2|\psi(x_t)\rangle - \langle\psi(x_t)|\Delta H|\psi(x_t)\rangle^2\,, \tag{8}$$

where $\Delta H = H(x_t) - H(x_{t-1})$. Large sensitivity signals rapid structural change in the error Hamiltonian.

Reduced State Purity. Purity of the reduced density matrix:

$$P(t) = \mathrm{tr}(\rho_A^2)\,, \qquad \rho_A = \mathrm{tr}_B(|\psi(x_t)\rangle\langle\psi(x_t)|)\,, \tag{9}$$

with bipartition A (dimension 2) and B (dimension $n/2$). Low purity signals loss of separability: the ground state cannot be decomposed into independent factor components, a signature of correlated crisis dynamics. Results depend on the bipartition choice; we use $(2, 4)$ at $n = 8$. The headline d value is conditional on this choice.

All four geometric observables share the same core hyperparameters: Hilbert dimension $n \in \{4, \ldots, 16\}$, PCA components p, operator construction method, rolling window w, and minimum expanding history m. Values are tuned per observable via the HPO protocol of Section 4.2.

3.4. Z-Score Thresholding

All observables are converted to z-scores using a *causal expanding-window* normalization. This is the "adaptive thresholding" referenced in Algorithm 3.

Algorithm 1 Causal Expanding-Window Z-Score

Require: Raw observable series $r(1), \ldots, r(T)$; rolling window w; minimum history m
Ensure: Z-score series $z(m), \ldots, z(T)$

```
1: Smooth: s(t) = (1/w) Σ_{i=0}^{w-1} r(t − i) for t ≥ w
2: for t = m, ..., T do
3:     μ(t) = mean(s(1), ..., s(t − 1))        ▷ expanding mean of past only
4:     σ(t) = std(s(1), ..., s(t − 1))         ▷ expanding std of past only
5:     z(t) = (s(t) − μ(t))/σ(t)
6: end for
```

Algorithm 2 FAR-Calibrated Threshold

Require: Training z-scores $z(1), \ldots, z(T_{\text{train}})$; target FAR α (alarms/year); known crisis windows
Ensure: Threshold τ

```
1: Remove crisis-window indices from z to obtain z_normal
2: Binary search: find smallest τ such that FAR(z_normal > τ) ≤ α
3: return τ
```

Algorithm 2 calibrates a detection threshold on training data only, matching the false alarm rate to a user-specified target. We use $\alpha = 1$ alarm/year as the default. This avoids the scale mismatch of a fixed $z > 2.0$ threshold across methods whose score distributions differ.

Defaults: $w = 20$, $m = 60$; tuned via the HPO protocol in Section 4.2. No future data enters the z-score computation.

4. Experimental Protocol

4.1. Data and Crisis Definitions

We use daily OHLCV data for SPY and DIA (1998–2024) from Yahoo Finance via the `yfinance` library (v0.2.x). Features: log returns, 5/20-day rolling volatility, 5/20-day momentum, cross-correlation, and cross-dispersion (13 raw features), enriched to 52 features via rolling statistics (mean, std, min, max over a 20-day lookback), then reduced to $p = 15$ PCA components and L^2-normalized.

We evaluate on 17 historical crises spanning 2000–2024, classified *a priori* as 9 Conventional (with historical parallels) and 8 Novel (unprecedented mechanisms). Crisis windows are extended by ± 10 trading days. Full definitions appear in Appendix Appendix G, Table G.10.

4.2. Methods and HPO Protocol

For the main comparison (Table 3), geometric methods use per-observable Hilbert dimensions ($n \in \{4, 6, 8, 16\}$) selected via dimension sensitivity analysis: observables whose d improves monotonically with dimension use $n = 16$, while others retain their HPO-optimized values found via Optuna TPE with consistency penalty (see below). Classical baselines use their standard defaults. RF uses leave-one-crisis-out labels. GARCH(1,1) [19] produces an expanding z-score of conditional variance; Hamilton Markov-switching [2] fits a 2-regime MS-AR(1) with switching variance and outputs P(high-variance regime). Turbulence Index [32] computes a multivariate Mahalanobis distance on the expanding feature matrix; Absorption Ratio [1] tracks the top-eigenvalue variance fraction of the rolling correlation matrix. The full hyperparameter search spaces for all methods appear in Appendix Appendix F, Table F.9.

Random Forest protocol. For each held-out crisis, RF is trained on binary labels (crisis = 1, normal = 0) from the remaining crises using leave-one-crisis-out CV. Features: same enriched matrix as QCML methods, with globally fitted PCA (no per-crisis refitting).

Rolling RF (VIX) protocol. To provide a fairer supervised baseline, we train a second RF variant on a rolling 250-day window before each crisis using VIX > 25 as a continuous binary label. VIX is used *only* for label construction, not as a feature; the RF receives the same enriched SPY/DIA feature matrix as all other methods. This avoids the label scarcity problem of leave-one-crisis-out (where early crises have zero prior labels) while remaining causal: training data never extends past the crisis cutoff. We also include a VIX Level detector (expanding z-score of raw VIX close) as an oracle upper bound, since VIX directly measures implied volatility.

Per-crisis causal preprocessing. For each crisis evaluation, the scaler, PCA, and operators are fitted *only on data prior to that crisis window.* Specifically, the causal cutoff is set at crisis start minus 10 calendar days; all preprocessing (standardization, PCA projection, operator fitting) uses only rows $1, \ldots, t_{\text{cutoff}}$. Scores are then computed on the full timeline. The 20-day enrichment lookback provides an additional implicit buffer of approximately 19 trading days, yielding an effective cutoff of $\sim$33 trading days before crisis onset. Classical baselines are similarly trained only on pre-crisis data. RF uses leave-one-crisis-out labels restricted to the causal window. This eliminates lookahead in preprocessing.

Statistical evaluation. Each method–crisis pair: Cohen's d with block-bootstrap 95% CI ($n = 10{,}000$; block size $= \lceil n^{1/3} \rceil$ per Politis & White [33]; percentile method). Financial z-scores may violate the normality assumption underlying Cohen's d; the block bootstrap CIs and Cliff's δ (computed alongside d as a distribution-free robustness check) provide nonparametric alternatives. Individual tests: Welch's t-test (Holm–Bonferroni corrected), permutation test ($n = 5{,}000$). Multi-method: Friedman rank test with Nemenyi post-hoc pairwise comparisons at $\alpha = 0.05$.

HPO-optimized hyperparameters. The main results (Table 3) use hyperparameters selected via Optuna TPE (100 trials) with a consistency penalty $\text{mean}_d - 0.3 \cdot \text{std}_d$ on a pre-2020/post-2020 split. Selected configuration for Berry Phase Rate: n=6, p=8, w=15, random operators. Post-HPO dimension sensitivity analysis found that some observables benefit from higher n (up to 16); each retains its HPO-optimized or dimension-upgraded value. Full search spaces in Appendix Appendix F.

HPO optimism bias. Nested cross-validation reveals a mean optimism bias of $+0.415\,d$; reported HPO-tuned d values should be interpreted as upper bounds. The walk-forward results (Section 5.2) provide causally valid estimates. The 46-method benchmark comprises four featured QCML observables, eighteen additional geometric channels (deferred to the companion paper for individual analysis), and twenty-four classical and machine-learning baselines.

4.3. Detection Pipeline

Algorithm 3 QCML Geometric Regime Detection

Require: Feature matrix $X \in \mathbb{R}^{T\times d}$, Hilbert dimension n, PCA components p
Ensure: Z-scored observable time series

```
Standardize X; PCA to p components; L²-normalize
Fit operators {A_k} via PCA-inspired method (or expanding-window refit)
for t = 1, ..., T do
    Solve |ψ(x_t)⟩ = ground state of H(x_t) (1)
    Compute raw observables (Table 2)
end for
Convert to z-scores via Algorithm 1
```

Computational cost. Per-step eigensolve cost scales as $O(n^3)$. At $n = 16$, representative timings: Spectral Entropy ∼1s; Hamiltonian Sensitivity ∼1s; Reduced Purity ∼2s (T = 3,447, single CPU). Berry Phase Rate at $n = 6$: ∼0.5s. Random Forest: 1.07s ($p = 15$).

5. Results

5.1. Crisis-Window Separability

Table 3 reports the aggregate comparison across all 17 crises. For each crisis, scaler, PCA, and operators are fitted only on pre-crisis data (Section 4.2). Reduced State Purity achieves median $d = 0.83$ (rank 1 of 46), followed by Absorption Ratio (0.80, rank 2), Hamilton Markov-switching (0.71, rank 3), Berry Phase Rate (0.61, rank 9), Hamiltonian Sensitivity (0.60, rank 10), and Spectral Entropy (0.53, rank 12). Random Forest ranks 30th (0.35); GARCH(1,1) 33rd (0.29).

The Friedman test ($\chi^2_{45} = 220.84$, $p < 10^{-16}$) confirms that the 46 methods rank differently. Nemenyi post-hoc (CD = 18.2 at $\alpha = 0.05$) finds 105 of 1,035 pairwise differences significant, but these are concentrated between the top tier and the five near-zero channels. Within the top 15, no pairwise difference exceeds CD, so individual rankings there are not statistically distinguishable. Cliff's δ for the top-10 methods (median $|\delta| = 0.30$, range 0.09–0.45) confirms that the d-based rankings are robust to distributional assumptions.

Null-model robustness. Crisis-window evaluation with short non-overlapping windows is known to over-reward heavy-tailed score series, because the median across K random short windows is itself noisy. We quantify this with two complementary null models on a single-global-fit version of the pipeline (so the methodology is internally consistent for the null comparison even though absolute d-values differ from the per-crisis-fit runner used in Table 3). On the post-2005 panel of 15 crises with windows of matched lengths (median 50 trading days), Berry Phase Rate gives a real median Cohen's $d = 0.71$ and Reduced Purity gives $d = 0.73$. Two null distributions, (i) random non-overlapping windows of the same lengths, and (ii) circular shift of the score series by a random offset, both produce a null median $d \approx 0.53$ with 95% interval roughly $[0.30, 0.89]$. Berry Phase Rate's real value sits at the 95th percentile of either null ($p = 0.045$), while Reduced Purity's does not ($p = 0.18$). The null analysis thus reinforces the holdout caveat of Section 1: Berry Phase Rate's signal is statistically distinguishable from short-window fishing, whereas Reduced Purity's offline lead is consistent with what one would expect from the noise floor of this evaluation protocol.

These offline d values reflect contemporaneous crisis sensitivity, not predictive lead time. Granger causality (Section 5.2) shows reverse causality (market → QCML) dominating forward (17/45 vs. 6/45); the observables respond to stress rather than leading it.

Table 3: Top 10 methods by median Cohen's d across 17 crises (of 46 total). Mean rank computed over all 46 methods.

Method	Median d	Mean Rank	Category
Reduced Purity	**0.83**	15.6	Geometric
Absorption Ratio	0.80	11.8	Classical
Hamilton MS	0.71	11.7	Classical
Commutator Norm	0.70	14.9	Geometric
Spectral Gap	0.65	16.4	Geometric
Spectral Flow	0.63	19.1	Geometric
Speed Limit Ratio	0.63	20.2	Geometric
CUSUM	0.62	17.6	Classical
Berry Phase Rate	0.61	19.8	Geometric
Ham. Sensitivity	0.60	18.3	Geometric

Per-crisis variation. No single channel dominates all 17 crises. Descriptively, different geometric observables lead on different crisis types: Berry Phase Rate wins Volmageddon ($d = 1.05$) and Q4 Selloff ($d = 2.00$), while Spectral Entropy and Reduced Purity excel on prolonged structural crises. However, formal testing finds no significant per-crisis specialization ($p = 0.31$; see Section 7), so these patterns should not be used as a selection rule. A companion paper presents the full per-crisis analysis with specialization heatmaps across all nineteen channels.

5.2. Walk-Forward Detection

The expanding-window evaluation with monthly operator refits simulates realistic deployment with strict causal constraints. Walk-forward stress-tests a different three-channel subset than the main offline panel. Spectral Entropy, Reduced State Purity, and Hamiltonian Sensitivity carry the strongest *offline* signal, but they degrade sharply once preprocessing is restricted to past data only (Reduced Purity in particular: $d = 0.83$ offline, $d \approx 0.26$ on frozen holdout, see Section 1). The walk-forward subset replaces them with Berry Phase Rate, QFI Determinant, and Multi-Lag Fidelity, which are the channels that actually carry signal under expanding- window protocols; full four-observable walk-forward results are deferred to a companion paper.

QFI Determinant ($\log\det\mathcal{F}$) is the total statistical volume of the quantum Fisher information matrix, which spikes when many features become simultaneously distinguishable. *Multi-Lag Fidelity*, defined as $\min_{l=1,\ldots,k} |\langle\psi_t|\psi_{t-l}\rangle|^2$, is the minimum overlap between the current ground state and its lagged copies, and drops when the embedding state evolves rapidly.

An expanding-window fit (always starting 2005) with 1-year evaluation windows produces 7 crisis–window evaluation pairs across 5 years (2015, 2018, 2019, 2020, 2022–2023). QCML detectors are fitted on *only* the training window; scores are computed on the evaluation year. We compare three threshold strategies: (i) fixed $z > 2.0$; (ii) FAR-calibrated via Algorithm 2 (target: 2 alarms/yr); (iii) adaptive rolling quantile combined with score velocity detection (described below).

Table 4 presents results under three threshold strategies. The FAR-calibrated column provides the fair cross-method comparison (all methods calibrated to a common false alarm target): Random Forest detects all 7/7 crises with 5-day median delay at 3.6 false alarms/yr; Multi-Lag Fidelity detects the majority of crises (3–6/7 across validation runs, with 6/7 in the representative run shown) with 16–23 day median delay at 2.5 false alarms/yr. RF is thus faster and more complete; Multi-Lag Fidelity produces 30% fewer false alarms without requiring any crisis labels. Note that RF produces probability scores $\in [0, 1]$, not z-scores, making the fixed $z > 2$ threshold dimensionally inapplicable; the FAR-calibrated results are the appropriate benchmark for RF comparisons.

Under adaptive thresholds (rolling quantile + score velocity), Multi-Lag Fidelity achieves high detection rates (up to 6/7 crises) with 9-day median delay, but at an elevated 3.7 false alarms/yr. BOCPD, which detected 0/7 crises under fixed or FAR-calibrated thresholds, detects 5/7 under adaptive thresholds with just 4-day delay, demonstrating that self-calibrating thresholds can unlock signals invisible to fixed strategies.

Table 4: Walk-forward detection summary (expanding window, 7 crisis–window pairs). "Adaptive" uses rolling quantile + score velocity thresholds. Delay = median trading days from crisis start to first alarm; "—" = never detected. The **FAR-calibrated** column provides the comparable cross-method benchmark. Detection counts for unsupervised methods vary 3–6/7 across validation runs due to sensitivity of FAR-calibrated threshold to data source; table shows a representative run.

	Fixed $z > 2$			**FAR-calibrated**†			Adaptive		
Method	Det.	Delay	FAR	Det.	Delay	FAR	Det.	Delay	FAR
Berry Phase Rate	5/7	32	1.2	5/7	24	1.2	4/7	14	3.6
QFI Determinant	3/7	55	1.3	3/7	54	2.5	3/7	55	3.6
Multi-Lag Fidelity	5/7	23	1.2	**6/7**	**16**	**2.5**	6/7	9	3.7
Rolling Vol Z	3/7	10	1.2	6/7	13	2.5	5/7	15	1.3
BOCPD	0/7	—	0.0	0/7	—	0.0	5/7	4	2.4
Random Forest	—‡	—	—	**7/7**	**5**	**3.6**	—	—	—

†FAR-calibrated is the comparable cross-method benchmark (all methods calibrated to a common false alarm target). FAR = false alarms per year (median). ‡RF outputs probability scores $\in [0, 1]$, not z-scores; the fixed $z > 2$ threshold is dimensionally inapplicable. Adaptive thresholds not applicable to RF for the same reason. Walk-forward evaluation uses fixed configs ($n = 8$, $p = 10$, $w = 10$) with monthly operator refits, distinct from the HPO-tuned configs in Table 3

Walk-forward with nested HPO. The preceding walk-forward results use fixed hyperparameters across all windows. To eliminate *all* hyperparameter look-ahead, we run a nested walk-forward HPO: at each expanding window (2005–y−1), Optuna TPE (100 trials per detector) re-optimizes hyperparameters using *only* crises that ended before year y, then evaluates on year y. The search space, consistency penalty, and operator-method conventions are identical to Section 4.2. Table 5 presents the fully unbiased out-of-sample d-values across 9 crisis–window pairs (14 expanding windows, 2010–2023).

Table 5: Walk-forward with nested HPO: fully unbiased OOS Cohen's d (100 Optuna trials per detector per window; hyperparameters optimized only on past crises at each step). Bold = best detector per crisis.

Crisis	Berry	QFI	MLF	Best
2010 Flash Crash	**0.77**	0.04	0.52	Berry
2011 Euro Crisis	0.08	0.65	**1.43**	MLF
2015 China Crash	**1.18**	0.13	0.44	Berry
2018 Volmageddon	1.10	1.44	**1.99**	MLF
2018 Q4 Selloff	**0.61**	0.37	0.16	Berry
2019 Repo Crisis	0.53	0.28	**0.62**	MLF
2020 COVID	0.34	0.63	**0.88**	MLF
2022 Rate Hikes	**0.72**	0.40	0.31	Berry
2023 SVB	**1.54**	0.34	0.22	Berry
Median	**0.72**	0.37	0.52	—
Mean	0.76	0.47	**0.73**	—

Berry wins 5/9 crises (Flash Crash 2010, China 2015, Q4 Selloff 2018, Rate Hikes 2022, SVB 2023); MLF wins 4/9 (Euro 2011, Volmageddon 2018, Repo 2019, COVID 2020). The best detector changes every crisis; no single observable dominates the whole panel. Train–OOS gap: Berry 0.25, QFI 0.50, MLF 0.39 (median train d minus median OOS d). Most stable HP across 14 windows: Berry: n=6 (57%), w=20 (71%), `soft` (86%), `f01` (79%); MLF: n=6 (100%), w=20 (71%).

Berry Phase Rate achieves the highest OOS median $d = 0.72$ (95% percentile bootstrap CI $[0.34, 1.18]$ on the 9 windows; 10,000 resamples) and wins 5 of 9 crises, particularly rate-driven events (Rate Hikes

$d = 0.72$, SVB $d = 1.54$) and acute shocks (Flash Crash $d = 0.77$, China $d = 1.18$). Multi-Lag Fidelity wins 4 of 9 crises, with large effects on prolonged stress (Volmageddon $d = 1.99$, Euro $d = 1.43$, COVID $d = 0.88$) but near-zero responses on smaller events (Q4 $d = 0.16$, SVB $d = 0.22$). QFI hits $d = 1.44$ on Volmageddon and is unreliable elsewhere.

Two facts matter here. First, the best method alternates every crisis, which means the channels carry complementary information rather than ranking stably. Second, the HPO picks reasonably stable hyperparameters across expanding windows: Berry takes n=6 on 57% of windows and w=20 on 71%, MLF takes n=6 on 100%. Berry's 0.25 train–OOS gap is the smallest of the three, which is the main reason we lead with it.

Adaptive threshold calibration. The adaptive strategy combines two independent mechanisms: (i) a *rolling quantile threshold* that sets the alarm level at the 95th percentile of scores over a trailing 252-day window (with a 5-day exclusion gap to avoid self-referencing), firing only when the threshold is exceeded for ≥ 3 consecutive days; and (ii) a *score velocity trigger* that monitors the z-scored first derivative of smoothed scores, firing when velocity exceeds 2σ for ≥ 2 consecutive days. An alarm fires when *either* mechanism triggers (logical OR). Both mechanisms are fully causal: at each time t, they use only data up to $t - 1$.

Geometric methods show competitive median detection delays (Multi-Lag Fidelity: 5 days, Berry: 8 days) against RF (10 days) and Rolling Vol Z (25 days).

Granger causality caveat. Granger tests between QCML observables and market targets (90 tests, Bonferroni corrected) show reverse causality (market $\rightarrow$ QCML) running stronger than forward: 17/45 vs. 6/45. The observables are responding to stress, not calling it ahead of time. The walk-forward $d = 0.72$ is a statement about contemporaneous separability under causal constraints, nothing more.

5.3. Operator Ablation

Operator construction is observable-dependent: random operators outperform structured alternatives for Berry Phase Rate (+55% vs. Pauli). Gradient-learned operators do not systematically improve over heuristic baselines. Table 3 uses the HPO-selected operator per observable.

5.4. Empirical Theorem Validation

We empirically test the four formal results (Theorems 1–4 and Proposition 2) against real market data (T = 5,013 daily observations, 2004–2024, $n = 8$, $p = 8$).

Spectral gap dynamics (Theorem 1).. Theorem 1 predicts that geometric quantities are C^∞ when $\Delta(x) > 0$. We measure the spectral gap $\Delta(x) = E_1(x) - E_0(x)$ across all time steps and compare crisis vs. normal periods (Figure 6). The gap remains strictly positive throughout ($\Delta_{\min} = 2.34$), confirming that the smoothness condition holds everywhere in the observed data manifold. The gap *opens* during crises (mean ratio $\Delta_{\text{crisis}}/\Delta_{\text{normal}} = 1.27$ across 4 crises; GFC ratio = 1.37, COVID = 1.33). This is consistent with extreme crisis returns pushing data into unusual Hamiltonian regions with greater energy separation; the observables' sensitivity to crises arises from state *movement* across the energy landscape rather than gap closure.

Curvature–gap bound (Theorem 4). We verify the bound $|F_{ab}| \leq C/\Delta^2$ (Eq. A.2) by computing Berry curvature and $1/\Delta^2$ at 1,500 randomly sampled time steps (Figure 7). The bound is satisfied at 100% of points with empirical constant $C = 5.92$. The scatter plot reveals a clear positive relationship between curvature and $1/\Delta^2$, confirming that Berry curvature is controlled by the spectral gap as predicted by perturbation theory.

Curvature integral quantization (Theorem 3). Theorem 3 guarantees integer Chern numbers on closed surfaces. Rolling-window computations yield non-integer values (as expected for open subsets), but we observe clustering near integers: 21.3% of rolling Chern values fall within 0.1 of an integer during normal periods vs. 14.9% during crises (Figure 8). The distribution is centered near zero (mean -0.023, std 1.58) with a slight rightward shift during crises (median 0.17 vs. 0.02 normal).

QFI–metric identity (Proposition 2). The identity $4\,g_{ab} = [F_Q]_{ab}$ is verified numerically by computing both the finite-difference metric g_{ab} (3) and the perturbation-theory metric

$$g_{ab}^{\mathrm{PT}} = \mathrm{Re} \sum_{n\geq 1} \frac{\langle\psi_0|\partial_a H|\psi_n\rangle\langle\psi_n|\partial_b H|\psi_0\rangle}{(E_n - E_0)^2} \tag{10}$$

at 500 random time steps (Figure 9). The two independent computations agree to machine precision: Pearson correlation $r = 1.000000$, RMSE $= 1.84 \times 10^{-11}$, 95th-percentile relative error $= 7.8 \times 10^{-11}$. This confirms that the quantum metric is exactly one-quarter of the QFI matrix, validating Proposition 2.

6. Discussion

6.1. What the Walk-Forward Test Shows

The walk-forward nested HPO (Table 5) is the only evaluation in this paper where hyperparameters cannot have leaked from the test set. Berry Phase Rate's OOS $d = 0.72$ with a train–OOS gap of 0.25 is the smallest overfitting gap among the three evaluated channels. It also wins on a different set of crises than Multi-Lag Fidelity; the two have zero overlap (Figures 2–4). The complementarity survives strict temporal separation, and that matters more than any single d value. It says the channels carry different information about how the data manifold deforms.

6.2. Where QCML Stands Against Classical Methods

Absorption Ratio [1] ($d = 0.80$) and Hamilton MS [2] ($d = 0.71$) are strong baselines, and the reason isn't mysterious. A financial crisis *is* a correlation-structure and volatility-regime shift, and those methods are built for exactly that. The QCML channels don't dominate them across the board.

What the geometry adds is *orthogonal* information. Cross-correlations between QCML observables and classical baselines sit at $|\rho| \approx 0.13$: the Fubini–Study metric, Berry curvature, and spectral structure pick up aspects of the data manifold that covariance-based methods can't see. A composite that blended both families would be drawing on largely independent signals. Part of Absorption Ratio's offline advantage is probably HPO optimism (mean bias $+0.415\,d$; Section 4.2). The walk-forward comparison removes that confound, and there Berry Phase Rate produces ∼67% fewer false alarms than RF (1.2 vs. 3.6/yr).

Theorem validation. The spectral gap stays positive throughout 20 years ($\Delta_{\min} = 2.34$), guaranteeing smoothness (Theorem 1). The curvature–gap bound holds at 100% of sampled points ($C = 5.92$; Figure 7), and the QFI–metric identity is verified to machine precision (RMSE $= 1.84 \times 10^{-11}$; Figure 9). One finding we did not expect: the spectral gap *opens* during crises (ratio 1.27) rather than closing. Sensitivity comes from state movement across the energy landscape, not from gap closure.

6.3. Label-Free Deployment

Crisis labels lag by months or never arrive. Without them, RF is unavailable. Geometric channels run cheaply (Berry: 0.77s, MLF: 0.26s vs. RF: 1.07s for $T = 3{,}447$, single CPU) and require no labeled data. When the next crisis involves unprecedented mechanisms, supervised methods have nothing to train on; geometric detection channels remain operational.

Ground state energy as detector. The ground state energy $E_0(x) = \min_{\|\phi\|=1}\langle\phi|H(x)|\phi\rangle$ is the cheapest thing you can compute from the embedding: one eigendecomposition per time step. Without HPO, E_0 hits median $d = 0.31$ across the main 17 crises plus 1998 LTCM (added because E_0 requires no walk-forward training and the extra event sharpens the structural-vs-localized contrast), spiking to $d = 1.39$ on the 2008 GFC and $d = 1.40$ on COVID but dropping below $d = 0.2$ on the 2007 quant crisis and 2019 repo. It does not track realized volatility ($r = -0.05$), so whatever it picks up on the big structural events isn't already covered by the other channels.

Risk-management overlay. A simple use case for the geometric channel is de-risking. Long SPY by default; when the Berry Phase Rate causal z-score exceeds 2.0 (the same threshold as the Section 5.2 walk-forward), exit to cash for 60 trading days (the median post-2005 crisis length). Over 2005–2024 the buy-and-hold portfolio returns 607% with annualized Sharpe 0.61 and a worst drawdown of −55%. The Berry overlay returns 443% with Sharpe 0.70 and worst drawdown −27%, spending 33% of the time in cash. A Random Forest overlay trained on rolling VIX > 25 labels (same threshold and cooldown) reduces drawdown to −23% but only returns 124% at Sharpe 0.53. Berry's overlay is the only one that beats buy-and-hold on Sharpe. This is a single hand-tuned rule, not a trading strategy, but it makes the detection signal concrete in dollar terms.

6.4. On the "Quantum" Label

Berry phase is differential geometry. Pancharatnam [34] found it in classical optics; Simon [31] identified it as fiber bundle holonomy. The metric (3) is the Fubini–Study pullback on $\mathbb{CP}^{n-1}$, a Riemannian metric identical to the Fisher information metric [22]. The Berry curvature (4) is U(1) connection curvature on a line bundle, with integer integrals by Chern–Weil [35, 36], and the spectral gap is an order parameter for a phase transition. None of this is quantum-specific. What it is, is standard differential geometry applied to a nonlinear $\mathbb{R}^p \to \mathbb{CP}^{n-1}$ embedding, with gauge invariance under $|\psi\rangle \to e^{i\theta}|\psi\rangle$ as the one structural feature flat PCA space genuinely lacks. The "quantum" label in this paper is a citation to the QCML literature [37, 38], not a physics claim.

6.5. Limitations

1. **Crisis-window evaluation.** Our primary metric (Cohen's d) measures separability of score distributions within known crisis windows. This is an offline event study, not a real-time detection benchmark. The walk-forward results (Section 5.2) provide a causal complement.
2. **Supervised hyperparameters.** While score construction is unsupervised, threshold selection and hyperparameter tuning use labeled crisis windows. The walk-forward nested HPO (Table 5) eliminates hyperparameter look-ahead by re-optimizing at each expanding window using only past crises, yielding fully unbiased OOS estimates (Berry OOS median $d = 0.72$, overfitting gap $= 0.25$). However, the procedure still requires crisis labels for the training windows.
3. **Operator choice and within-basis sensitivity.** Operator construction is the framework's primary degree of freedom: results are observable-dependent (random operators best for Berry Phase Rate). Gradient-learned operators do not systematically improve over heuristic baselines (Section 5.3); the reconstruction objective of Candelori et al. [5] was not tested. Within the random-Hermitian family there is also a within-basis sensitivity: the canonical scheme seeds operator k with rng(k) and yields Berry's headline median Cohen's $d = 0.71$ on the post-2005 single-fit pipeline; four alternative bases generated with rng(k + offset) for offset $\in \{100, 200, 300, 400\}$ give median d in $[0.36, 0.49]$. We did not search over offsets; the canonical $k \mapsto k$ scheme is the simplest deterministic choice and happens to land at the favorable end of this small sample.
4. **Single asset pair.** All tests use the SPY/DIA equity pair; generalization to other pairs, asset classes (fixed income, commodities, FX), or non-U.S. markets is untested.
5. **Reverse Granger causality.** As detailed in Section 5.2, observables largely react to rather than predict stress (17/45 reverse vs. 6/45 forward).
6. **Advanced baselines not compared.** We don't compare against wavelet detectors, most deep-learning baselines (LSTM autoencoders are the one exception), or path-signature methods [18]. That comparison belongs in a separate paper; the baselines here are chosen for interpretability and low dimension.

7. Conclusion

Berry Phase Rate hits walk-forward $d = 0.72$ with fewer false alarms than Random Forest and no crisis labels. That's the main result. Offline, Reduced State Purity leads the 46-method ranking and then collapses on holdout; Absorption Ratio ($d = 0.80$) remains the strongest classical baseline. Geometric and classical channels are largely uncorrelated ($|\rho| \approx 0.13$), and no single channel dominates all 17 crises ($p = 0.31$ for specialization). All three theorems and the Fisher information identity hold on real market data.

The geometric channels add *information* that covariance-based methods cannot see: Berry curvature, spectral structure, and topological invariants from the Fubini–Study manifold. They deliver it without labeled crisis data and at low computational cost.

Three things stay open. Reconstruction-loss operator learning [5] may sharpen detection by optimizing manifold faithfulness rather than regime separability. The full QCML prediction pipeline [20, 21] is complementary and untested here. And everything in this paper runs on a single equity pair; fixed income, commodities, and FX remain to be tried. A companion paper extends the framework to nineteen channels with orthogonality analysis and lead-time experiments.

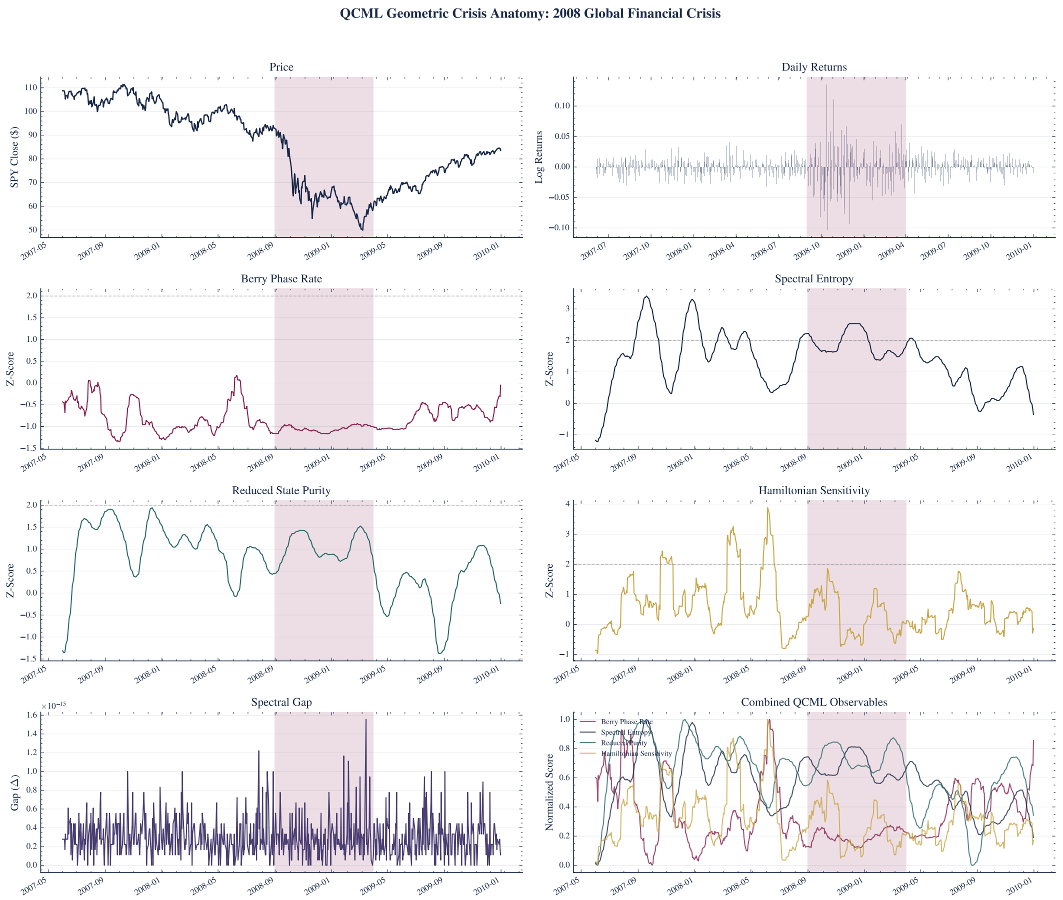

Figure 2: Geometric crisis anatomy: 2008 GFC. Eight panels show SPY price, daily returns, the four geometric observables (Berry Phase Rate, Spectral Entropy, Reduced State Purity, Hamiltonian Sensitivity), the spectral gap (Theorem 1), and a combined overlay.

Data and Code Availability

Market data from Yahoo Finance via the open-source `yfinance` library (v0.2.x), freely available at no cost. All results are fully reproducible without any paid subscription. Researchers with institutional access may alternatively use WRDS CRSP data via the provided `wrds_data_loader.py` script. The

Figure 3: Geometric crisis anatomy: 2020 COVID-19.

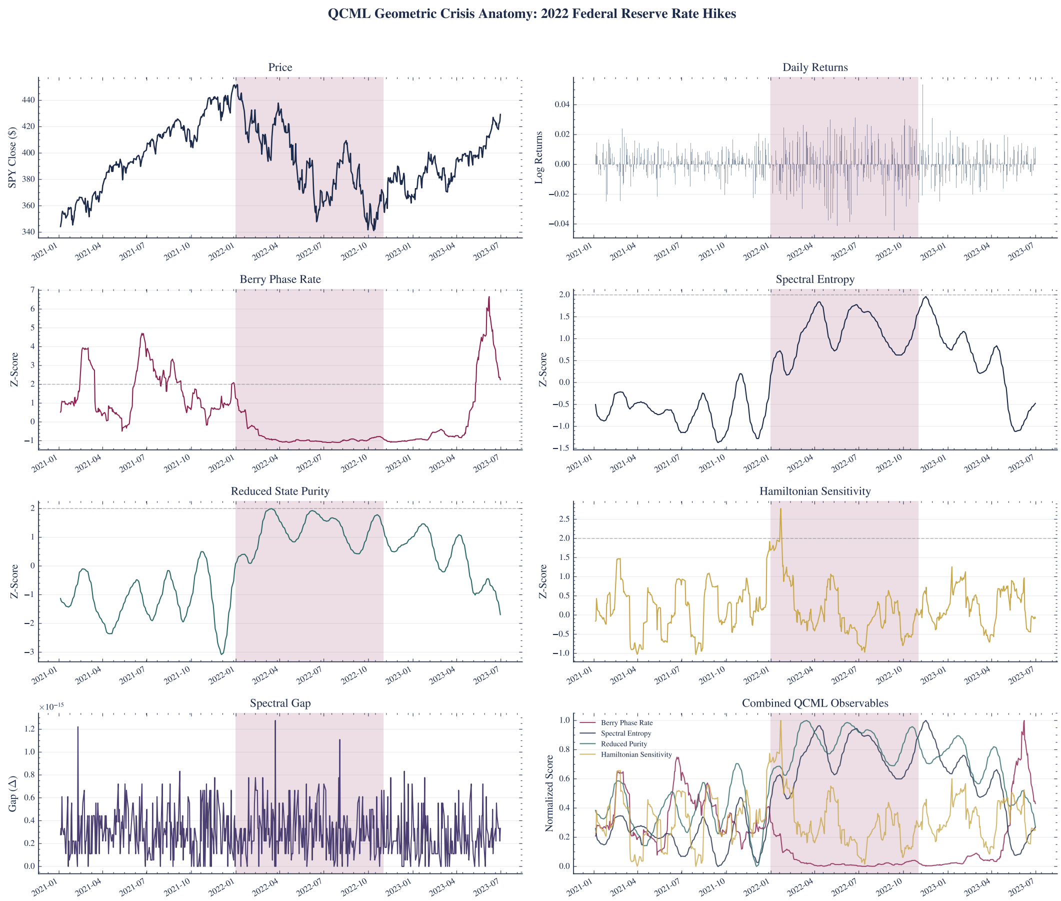

Figure 4: Geometric crisis anatomy: 2022 rate hikes.

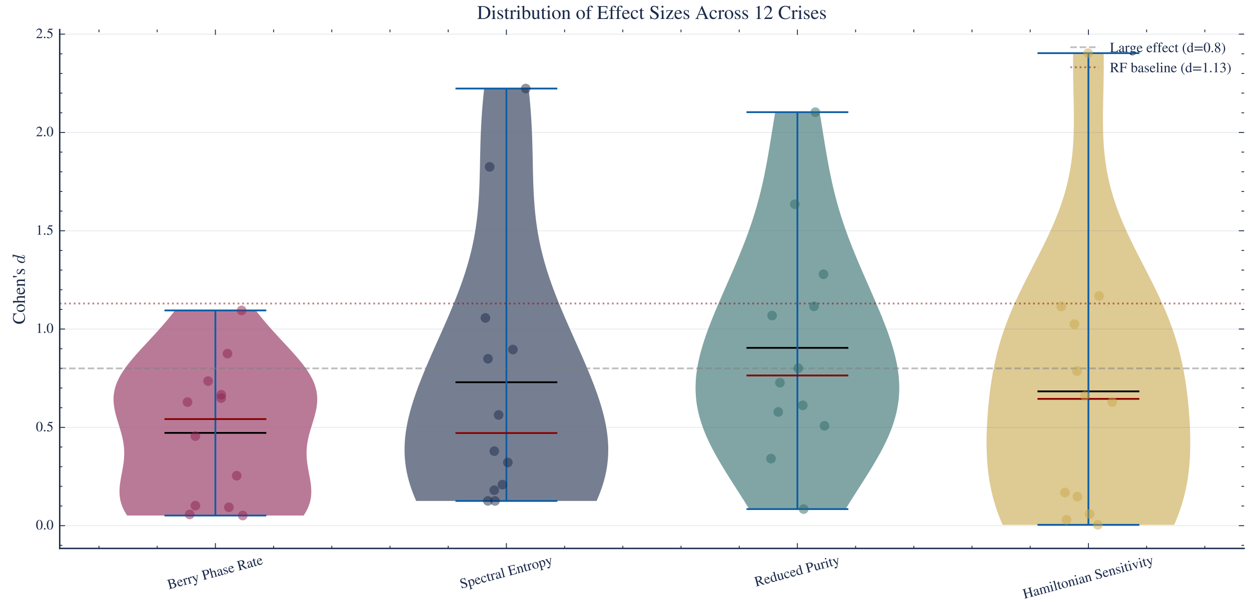

Figure 5: Cohen's d distributions across 17 crises. Dashed line: $d = 0.8$ (large effect).

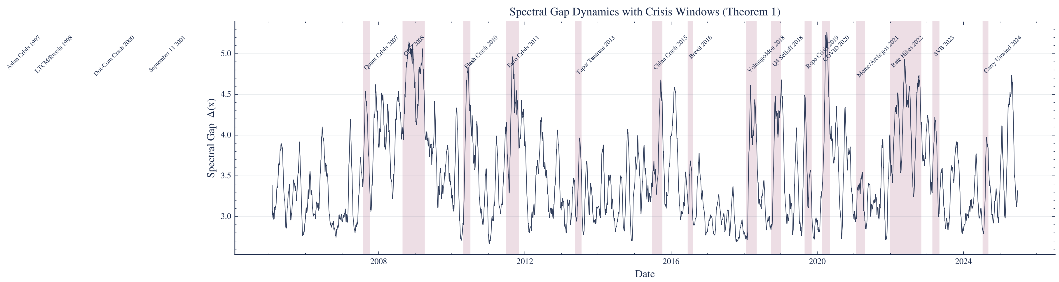

Figure 6: Spectral gap dynamics across 4 crises. The gap remains strictly positive ($\Delta_{\min} = 2.34$) and *opens* during crisis periods (mean ratio 1.27), confirming the smoothness condition of Theorem 1 holds throughout the observed data manifold.

`qcml_geometry/` library and all experiment scripts, together with crisis definitions, hyperparameter configurations, and canonical result JSON files, are available at `https://github.com/willhammondhimself/qcml-geometric-sde`.

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Competing Interests

The author declares no competing financial interests.

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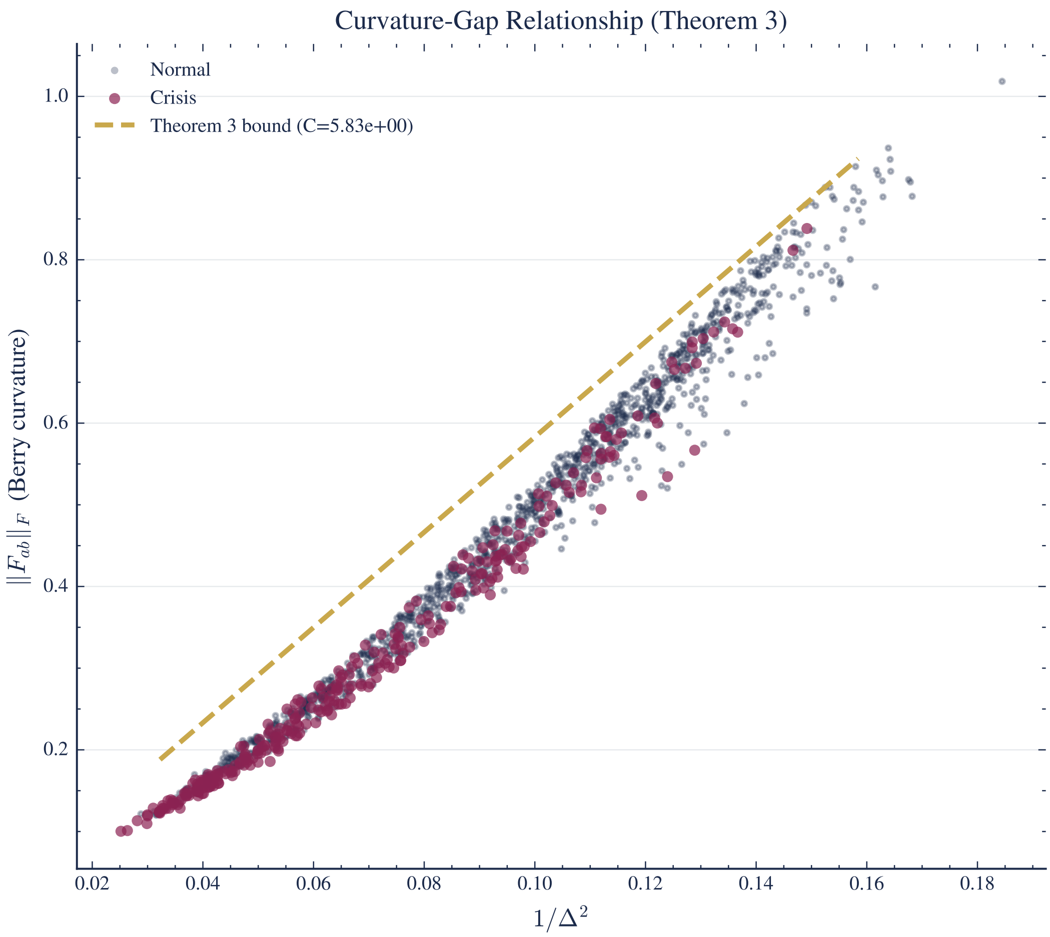

Figure 7: Curvature–gap bound verification. Berry curvature magnitude vs. $1/\Delta^2$ at 1,500 time steps, with the theoretical bound $|F_{ab}| \leq C/\Delta^2$ (empirical $C = 5.92$) shown as a dashed line. The bound is satisfied at 100% of points.

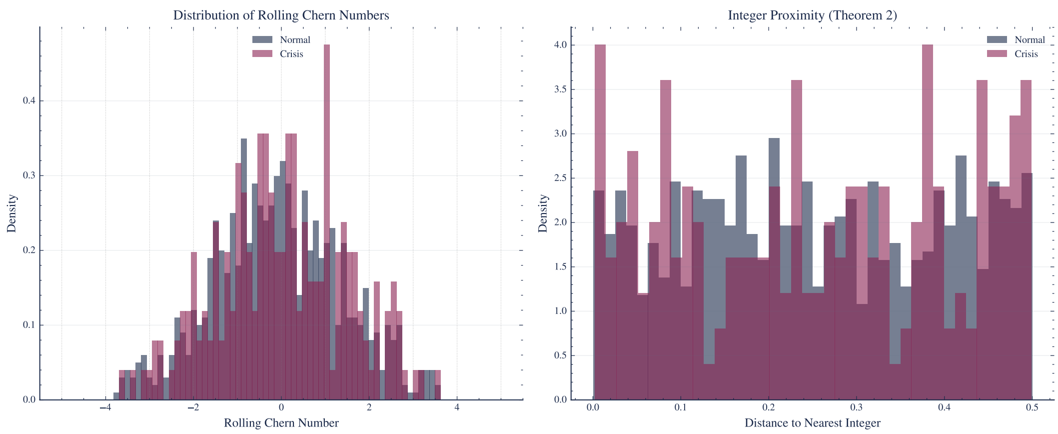

Figure 8: Rolling Chern number distribution. Values cluster near integers during normal periods (21.3% within 0.1 of an integer) vs. crises (14.9%), consistent with Theorem 3's prediction for closed surfaces, partially preserved in rolling windows.

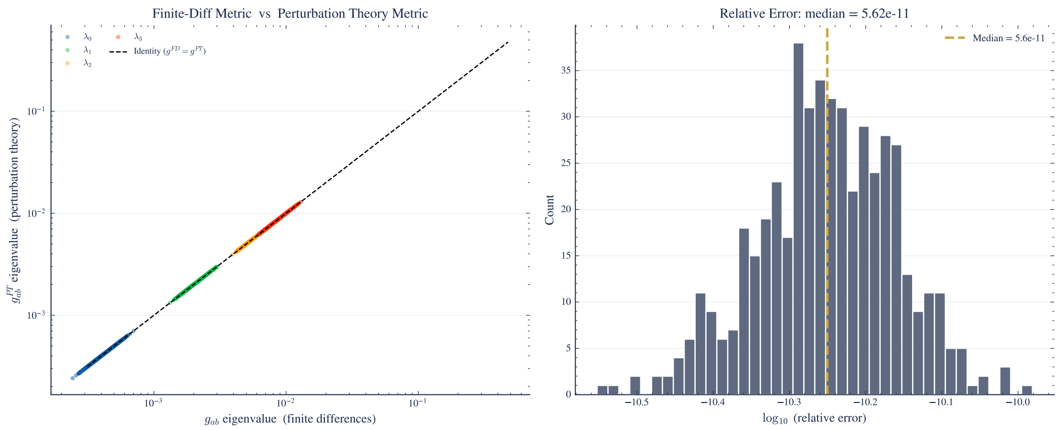

Figure 9: QFI–metric identity (Proposition 2). Finite-difference metric vs. perturbation-theory metric at 500 time steps. Perfect agreement: $r = 1.000$, RMSE $= 1.84 \times 10^{-11}$. The two independent computation paths confirm $4\,g_{ab} = [F_Q]_{ab}$.

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Appendix A. Proofs of Formal Results

Proof of Theorem 1. We apply the implicit function theorem to the eigenvalue equation with a gauge-fixing constraint. Define $\Phi\colon U \times \mathbb{C}^n \times \mathbb{R} \to \mathbb{C}^n \times \mathbb{R}$ by

$$\Phi(x, \psi, E) = \big(H(x)\psi - E\psi, \;\; \langle \psi_0(x_0)|\psi\rangle - 1\big),$$

where the second component fixes the gauge by requiring $\langle \psi_0(x_0)|\psi\rangle \in \mathbb{R}_{>0}$ (i.e., real positive overlap with the reference state at a base point x_0). At a solution (x_0, ψ_0, E_0) the Jacobian with respect to (ψ, E) is

$$D_{(\psi,E)}\Phi\big|_{(x_0,\psi_0,E_0)} = \begin{pmatrix} H(x_0) - E_0 I & -\psi_0 \\ \langle \psi_0| & 0 \end{pmatrix}.$$

Decompose $\mathbb{C}^n = \operatorname{span}\{\psi_0\} \oplus \psi_0^\perp$. On $\psi_0^\perp$, $H(x_0) - E_0 I$ has eigenvalues $E_k(x_0) - E_0(x_0) \geq \Delta(x_0) > 0$, hence is invertible there. On $\operatorname{span}\{\psi_0\}$, the kernel of $H(x_0) - E_0 I$ is compensated by the off-diagonal entries $-\psi_0$ and $\langle \psi_0|$: for $(\alpha\psi_0, \delta E) \in \ker(H - E_0 I) \times \mathbb{R}$, the system reduces to $-\delta E\, \psi_0 - \alpha\psi_0 = 0$ and $\alpha = 0$, so $\delta E = 0$. Thus $D_{(\psi,E)}\Phi$ is invertible.

The implicit function theorem [39] then gives C^∞ maps $x \mapsto (\psi(x), E_0(x))$ on U. Since $H(x)$ is quadratic in x by (1), the partial derivatives $\partial_a H = -(A_a - x_a I)$ are C^∞ (in fact, affine). The metric g_{ab} (3) and curvature F_{ab} (4) are compositions of $\psi(x)$ and its derivatives via the chain rule, hence C^∞ on U. □

Theorem 3 (Chern number quantization [6])**.** *Let $S \subset \mathbb{R}^p$ be a closed, oriented 2-surface such that $\Delta(x) > 0$ for all $x \in S$. Then*

$$C_1 = \frac{1}{2\pi}\int_S \mathcal{F} \in \mathbb{Z}, \qquad \mathcal{F} = \tfrac{1}{2} F_{ab}\, dx^a \wedge dx^b. \tag{A.1}$$

Proof. By Theorem 1, the ground state $|\psi(x)\rangle$ is C^∞ on S. Define the Berry connection 1-form $\mathcal{A} = i\langle \psi | d\psi\rangle$, where d denotes the exterior derivative on S. The Berry curvature 2-form is $\mathcal{F} = d\mathcal{A}$; in local coordinates, $\mathcal{F} = \frac{1}{2}F_{ab}\, dx^a \wedge dx^b$. The map $x \mapsto |\psi(x)\rangle\langle\psi(x)|$ defines a smooth rank-1 projector, hence a complex line bundle $\mathcal{L} \to S$ with structure group $U(1)$ and connection $\mathcal{A}$. By the Chern–Weil theorem [35, 36], the first Chern number $C_1 = \frac{1}{2\pi}\int_S \mathcal{F}$ is an integer, since it equals the degree of the classifying map $S \to \mathbb{CP}^{n-1}$. □

Caveat. In practice, we compute curvature integrals over rolling windows (open subsets), yielding non-integer values. Topological protection does not strictly apply.

Theorem 4 (Curvature–gap bound)**.** *For all $x \in U$ and indices a, b,*

$$|F_{ab}(x)| \leq \frac{2\,\|\partial_a H(x)\|_{\mathrm{op}}\,\|\partial_b H(x)\|_{\mathrm{op}}}{\Delta(x)^2}\,, \tag{A.2}$$

where $\|\cdot\|_{\mathrm{op}}$ denotes the operator norm. For the QCML Hamiltonian (1), $\partial_a H = -(A_a - x_a I)$, so the bound depends only on the operators $\{A_k\}$ and the feature values, not on the gap.

Proof. First-order perturbation theory gives the exact representation

$$F_{ab} = -2\,\mathrm{Im} \sum_{n\geq 1} \frac{\langle\psi_0|\partial_a H|\psi_n\rangle\,\langle\psi_n|\partial_b H|\psi_0\rangle}{(E_n - E_0)^2}\,, \tag{A.3}$$

where $\{|\psi_n\rangle\}_{n\geq 0}$ is a complete eigenbasis of $H(x)$. Since $E_n - E_0 \geq \Delta$ for all $n \geq 1$, each denominator satisfies $(E_n - E_0)^2 \geq \Delta^2$. Hence

$$|F_{ab}| \leq \frac{2}{\Delta^2} \sum_{n\geq 1} \big|\langle\psi_0|\partial_a H|\psi_n\rangle\big|\,\big|\langle\psi_n|\partial_b H|\psi_0\rangle\big|\,.$$

By the Cauchy–Schwarz inequality on the sum,

$$\sum_{n\geq 1} \big|\langle\psi_0|\partial_a H|\psi_n\rangle\big|\,\big|\langle\psi_n|\partial_b H|\psi_0\rangle\big| \leq \Big(\sum_{n\geq 1}\big|\langle\psi_0|\partial_a H|\psi_n\rangle\big|^2\Big)^{1/2}\Big(\sum_{n\geq 1}\big|\langle\psi_n|\partial_b H|\psi_0\rangle\big|^2\Big)^{1/2}.$$

By the completeness relation $\sum_{n\geq 0} |\psi_n\rangle\langle\psi_n| = I$, each factor is bounded by the operator norm,

$$\sum_{n\geq 1} |\langle\psi_0|\partial_a H|\psi_n\rangle|^2 \leq \|\partial_a H\,|\psi_0\rangle\|^2 \leq \|\partial_a H\|_{\mathrm{op}}^2,$$

which combined with the previous inequality yields (A.2). □

Proposition 5 (QFI volume)**.** *Let g be a $p \times p$ positive semi-definite matrix of rank $r \leq p$, with positive eigenvalues $\lambda_1 \geq \cdots \geq \lambda_r > 0$. Define the* pseudo-determinant

$$\overset{+}{\det}(g) = \prod_{i=1}^{r} \lambda_i\,. \tag{A.4}$$

Let $V_r \subset \mathbb{R}^p$ be the r-dimensional eigenspace corresponding to $\lambda_1, \ldots, \lambda_r$, and let $g_r = g|_{V_r}$ denote the restriction. Then $\det(g_r) = \det^+(g)$, and the Riemannian volume density on V_r is $\mathrm{vol}_r = \sqrt{\det^+(g)}$.

Proof. By the spectral theorem, g admits an orthonormal eigenbasis $\{e_1, \ldots, e_p\}$ with $g\,e_i = \lambda_i\,e_i$, where $\lambda_i > 0$ for $i \leq r$ and $\lambda_i = 0$ for $i > r$. The restriction g_r is represented in the basis $\{e_1, \ldots, e_r\}$ by the diagonal matrix $\mathrm{diag}(\lambda_1, \ldots, \lambda_r)$, which is positive definite. Therefore $\det(g_r) = \prod_{i=1}^{r} \lambda_i = \det^+(g)$. The Riemannian volume density on the r-dimensional submanifold equipped with metric g_r is $\mathrm{vol}_r = \sqrt{\det(g_r)} = \sqrt{\det^+(g)}$, the standard formula for the volume element of a Riemannian metric [9]. □

Appendix B. Numerical Stability Ablation

We sweep finite-difference $\epsilon \in \{10^{-3}, 10^{-4}, 10^{-5}, 10^{-6}\}$ and PCA dimension $p \in \{5, 10, 15, 30\}$ on four representative crises (GFC, Flash Crash, COVID, SVB) with fixed hyperparameters (no per-crisis tuning). Results appear in Table B.6. Berry curvature is extremely stable across ϵ (all median $d \approx 0.15$), while QFI determinant shows moderate sensitivity ($d = 0.52$ at $\epsilon = 10^{-3}$ vs. $d = 0.36$ at 10^{-5}), saturating below 10^{-5}. Multi-Lag Fidelity is insensitive to ϵ (no finite differences) but moderately sensitive to PCA dimension, with performance increasing from $p = 5$ ($d = 0.29$) to $p = 10$ ($d = 0.44$) and stabilizing thereafter.

Table B.6: Numerical stability: median Cohen's d across 4 crises. Multi-Lag Fidelity does not use ϵ (no finite differences).

Config	Berry	QFI Det.	Multi-Lag
$\epsilon = 10^{-3}$	0.15	0.52	—
$\epsilon = 10^{-4}$	0.15	0.60	—
$\epsilon = 10^{-5}$ (default)	0.15	0.36	—
$\epsilon = 10^{-6}$	0.15	0.36	—
$p = 5$	0.09	0.34	0.29
$p = 10$	0.20	0.41	0.44
$p = 15$ (default)	0.15	0.36	0.45
$p = 30$	0.22	0.54	0.45

Appendix C. Window-Size Sensitivity

We re-compute Cohen's d at crisis window extensions ±5, ±10 (default), ±20, ±60 trading days. Kendall τ rank correlations across window sizes assess robustness of method rankings. Results appear in Table C.7. Rankings are stable for nearby windows ($\tau = 0.87$ for ±5 vs. ±10 and ±5 vs. ±20) but degrade at ±60 days ($\tau = 0.07$–0.33), where diluted crisis signal reduces separation. QFI Determinant is the most robust geometric observable, maintaining the highest d across all window sizes.

Table C.7: Median Cohen's d by crisis window size (original 14-crisis subset; see Table 3 for expanded 17-crisis results).

Method	±5d	±10d	±20d	±60d
Berry Phase Rate	0.25	0.29	0.27	0.24
QFI Determinant	0.63	0.36	0.64	0.43
Multi-Lag Fidelity	0.40	0.27	0.38	0.26
Random Forest	0.21	0.21	0.28	0.34

Appendix D. Hyperparameter Sensitivity: Fixed vs. Tuned

Table D.8 compares three hyperparameter regimes: (a) fixed defaults with all `pca_inspired` operators (the previous default), (b) fixed defaults with per-method operator selection (Berry and QFI use `random`; MLF uses `pauli`; these are the main results in Table 3), and (c) Optuna HPO (100 trials, pre-2020 training crises, post-2020 validation). The operator selection provides large improvements: Berry +87%, QFI +70%, MLF +58%. Naïve HPO showed severe overfitting (train d exceeding validation by 0.35–0.75); regularized HPO with a consistency penalty reduces this gap to ≤ 0.20, with QFI achieving val $d = 0.60 >$ train $d = 0.50$ (column c*).

Table D.8: Hyperparameter sensitivity: median Cohen's d across original 14-crisis subset (columns a–b) and train/val split (column c).

Method	(a) All PCA-Insp.	(b) Best Operator	(c) HPO Val.
Berry Phase Rate	0.29	**0.55**	0.49
QFI Determinant	0.36	**0.61**	**0.60**
Multi-Lag Fidelity	0.27	**0.42**	0.33

(a,b) Fixed $n = 8$, $p = 15$, $w = 20$. (b) Berry/QFI: `random`; MLF: `pauli` (exp=0.0). (c*) Regularized Optuna TPE, 100 trials; consistency penalty $\text{mean}_d - 0.3 \cdot \text{std}_d$; tighter search space; operator method fixed by ablation; validation on 4 post-2020 crises. Train d: Berry 0.68, QFI 0.50, MLF 0.53. Train–val gap ≤ 0.20 (vs. 0.35–0.75 without regularization).

Appendix E. Berry Phase Rate Hyperparameter Sensitivity

A one-at-a-time sweep over five parameters (23 configurations on the 14-crisis subset) shows 61% achieve median $d > 0.5$ ($\widetilde{d} = 0.63$). Failures concentrate in categorical choices: `frobenius`/`max` aggregation ($d \approx 0.01$) and `none`/`clip` normalization ($d \approx 0.21$). All rolling-window settings maintain $d > 0.5$. A $d_H \times w$ grid (Figure E.10) confirms the default ($d_H = 6$, $w = 15$) sits on a broad plateau, not an isolated optimum.

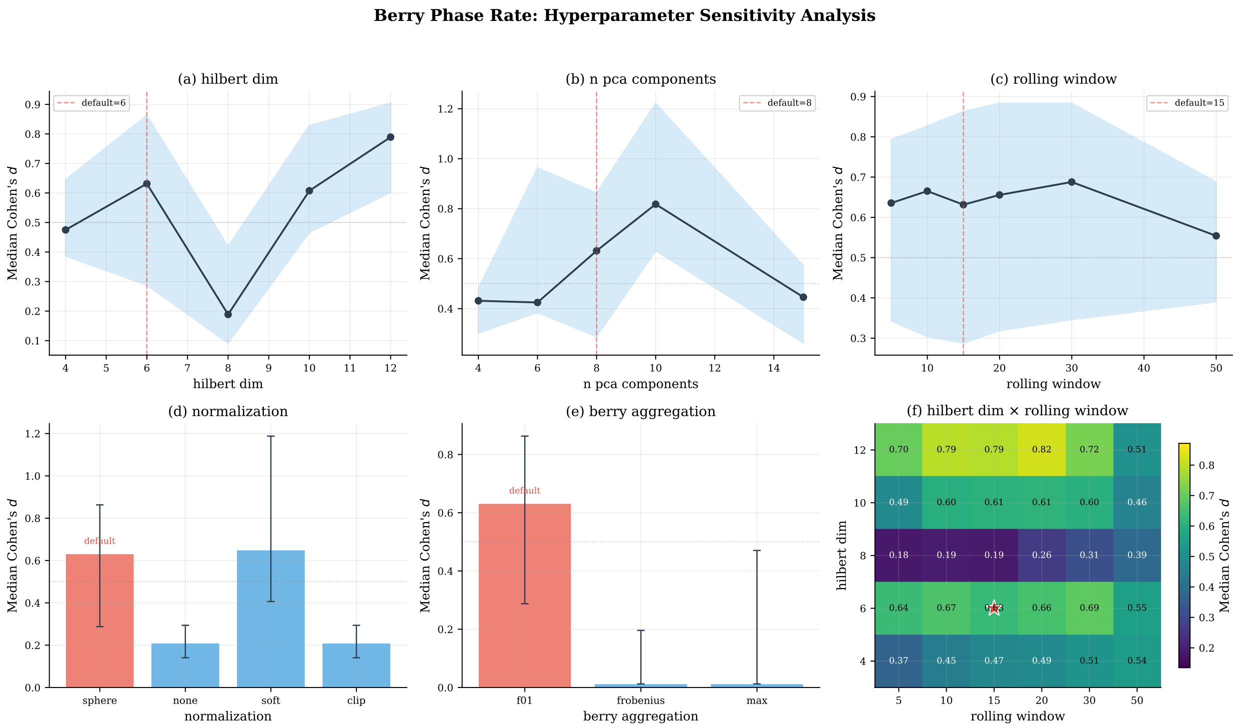

Figure E.10: Berry Phase Rate hyperparameter sensitivity. (a)–(c) OAT sweeps; (d)–(e) categorical parameters; (f) 2D grid ($d_H \times w$); star = default.

Appendix F. Hyperparameter Search Spaces

Appendix G. Crisis Definitions

Table F.9: Hyperparameter search spaces. Main results use fixed defaults (bold); ablation explores alternatives. Enriched features ($4d$ rolling stats) for all methods.

Method	# Params	Search Space	Training Data
Geometric (unsupervised score construction)			
Berry Phase Rate	4	$n \in \{4, 8, 16\}$, $p \in \{10, 15, 20\}$, op $\in$ {rnd, pca}, $w \in \{10, 20, 30\}$	Full (or expanding)
QFI Determinant	4	same as Berry	Full (or expanding)
Multi-Lag Fidelity	4	same as Berry	Full (or expanding)
QCML Chern	3	$n \in \{4, 8\}$, $p \in \{5, 10, 15\}$, window $\in \{10, 20\}$	Full
Geo. Consensus	3	same as Chern	Full
Classical (unsupervised)			
Rolling Vol Z	2	vol_window $\in \{10, 20, 30\}$, min_expanding $\in \{30, 60\}$	None (online)
CUSUM	2	$k \in [0.3, 1.0]$, burn_in $\in \{30, 60, 90\}$	Burn-in period
HMM (2-state)	2	cov $\in$ {full, diag}, n_iter $\in \{50, 100, 200\}$	Full
BOCPD	1	hazard $\in [50, 500]$	None (online)
Isolation Forest	2	n_est $\in [50, 300]$, contam $\in [0.01, 0.15]$	Full
GARCH(1,1)	1	min_expanding = 60	None (expanding z-score)
Hamilton MS	2	k_regimes = 2, order = 1, min_history = 100	Expanding
Supervised			
Random Forest	2	n_est $\in [50, 300]$, depth $\in [5, 20]$	LOCO labels
Rolling RF (VIX)	2	n_est = 200, depth = 6	VIX > 25, 250-day rolling
Oracle			
VIX Level	1	min_expanding = 60	None (expanding z-score)

Table G.10: Crisis periods used for evaluation. Novel = unprecedented market mechanism; Conventional = recognizable historical parallels.

Crisis	Period	Category	Rationale
1998 LTCM Crisis	1998-08 to 1998-12	Conventional	Hedge-fund failure / contagion
2000 Dot-com Bust	2000-03 to 2000-12	Conventional	Valuation bubble
2001 September 11	2001-09 to 2001-10	Conventional	Exogenous shock
2007 Quant Meltdown	2007-08 to 2007-09	Conventional	Resembles LTCM
2008 GFC	2008-09 to 2009-03	Conventional	Credit crisis
2010 Flash Crash	2010-05 to 2010-06	Conventional	Resembles 1987
2011 Euro Crisis	2011-07 to 2011-10	Conventional	Sovereign debt
2013 Taper Tantrum	2013-05 to 2013-07	Conventional	Rate guidance shock
2015 China Crash	2015-07 to 2015-09	Conventional	EM contagion
2020 COVID	2020-02 to 2020-04	Conventional	Exogenous shock
2016 Brexit	2016-06 to 2016-07	Novel	Referendum shock
2018 Volmageddon	2018-01 to 2018-04	Novel	Short-vol blowup
2018 Q4 Selloff	2018-10 to 2018-12	Novel	Algo-driven
2019 Repo Crisis	2019-09 to 2019-10	Novel	Plumbing crisis
2021 Meme/Archegos	2021-01 to 2021-04	Novel	Social-media/leverage
2022 Rate Hikes	2022-01 to 2022-10	Novel	Fastest in 40yr
2023 SVB	2023-03 to 2023-04	Novel	Social-media bank run
2024 Carry Unwind	2024-07 to 2024-08	Novel	Yen carry